

Large language models

NLP Week 8

Thanks to Dan Jurafsky and Marius Mosbach for some of the slides this week!

Plan for today

1. What are Large language models (LLMs)?
2. Pretraining and finetuning
3. Sampling and Decoding: How text generation works
4. Prompting: all tasks as text-to-text generation
5. Evaluating LLMs
6. *Group discussion: Impacts - Harms of LLMs and accountability*

Large language models

LARGE language models, LMs scaled up.

In practice: Computational agents that can interact conversationally with people using natural language

LLMs have revolutionized the field of NLP and AI

Fundamental intuition of LLMs (why scale?)

Fundamental intuition of LLMs (why scale?)

Text contains enormous amounts of knowledge

Pretraining on lots of text with all that knowledge is what gives language models their ability to do so much.

What does a model learn from pretraining?

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- With roses, dahlias, and peonies, I was surrounded by

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- With roses, dahlias, and peonies, I was surrounded by flowers
- The room wasn't just big it was _____

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What does a model learn from pretraining?

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- The square root of 4 is 2
- The author of "A Room of One's Own" is Virginia

What does a model learn from pretraining?

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What does a model learn from pretraining?

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- The author of "A Room of One's Own" is Virginia Woolf
- The professor told me that ____

What does a model learn from pretraining?

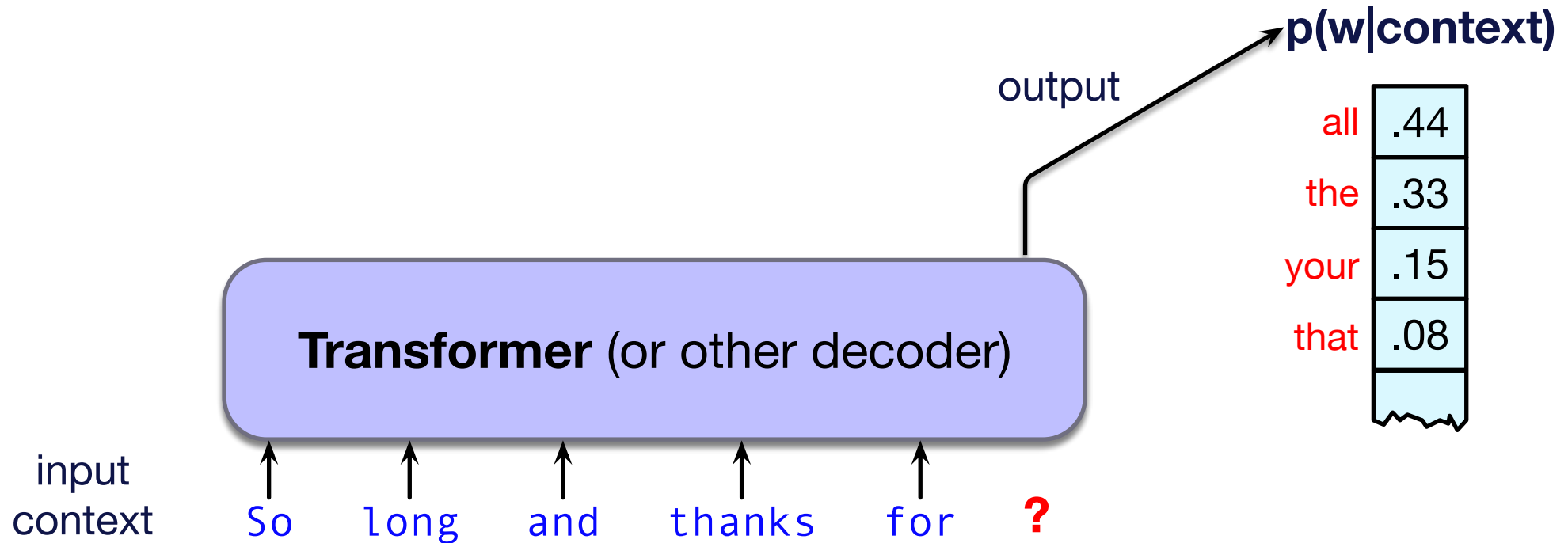
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What is a large language model?

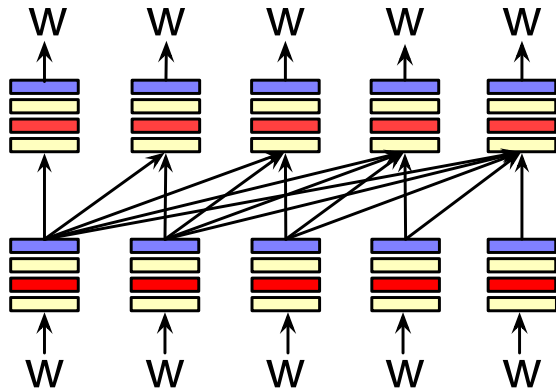
A neural network with:

Input: a context or prefix,

Output: a distribution over possible next words

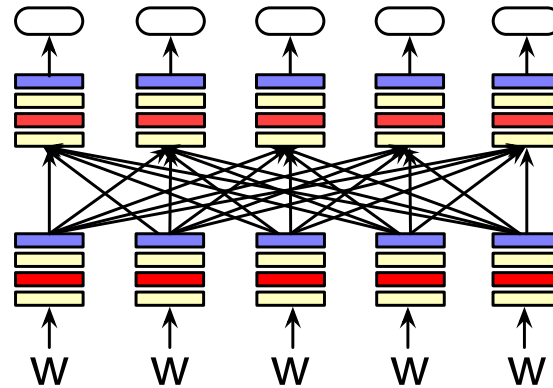


Three architectures for large language models



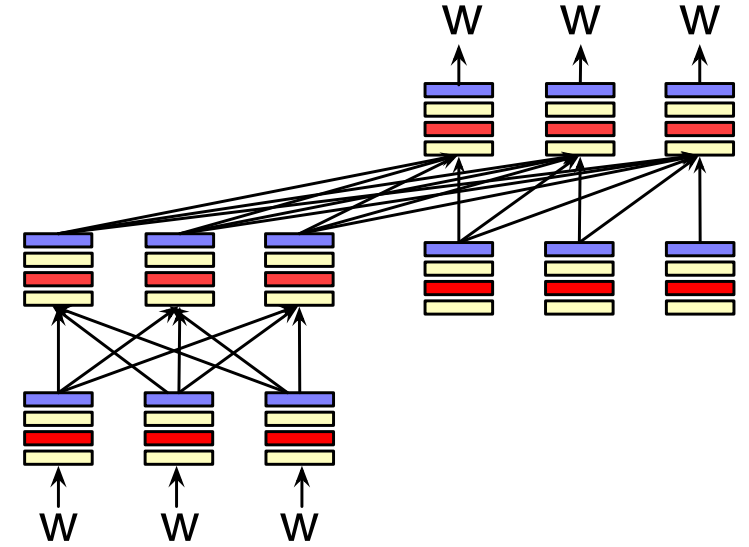
Decoders

GPT, Claude,
Llama
Mixtral



Encoders

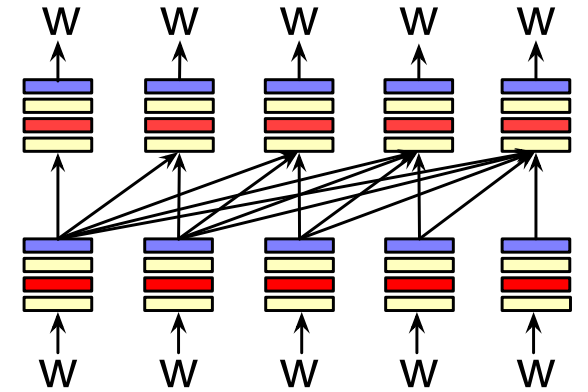
BERT family,
HuBERT



Encoder-decoders

Flan-T5, Whisper

Decoders

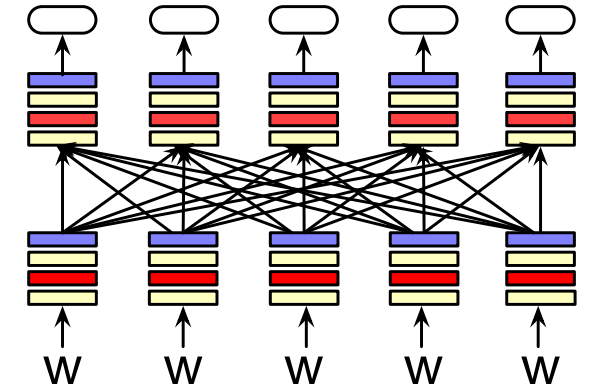


What most people think of when we say LLM

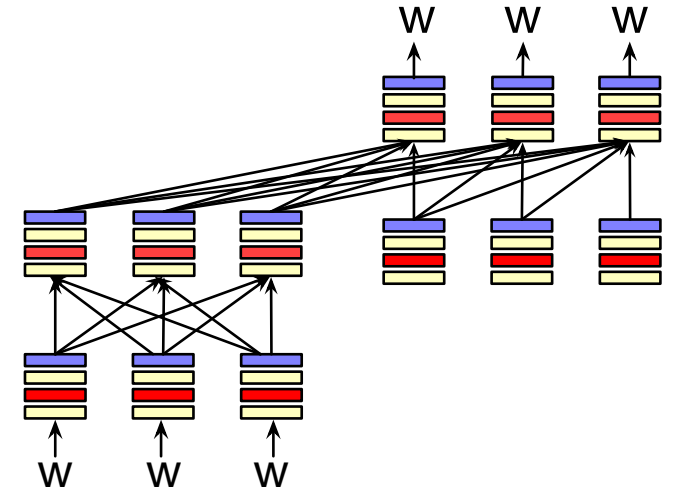
- GPT, Claude, Llama, DeepSeek, Mistral
- A generative model
- It takes as input a series of tokens, and iteratively generates an output token one at a time.
- Left to right (causal, autoregressive)

Encoders

- Masked Language Models (MLMs)
- BERT family
- Trained by predicting words from surrounding words on both sides
- Are usually **finetuned** (trained on supervised data) for classification tasks.



Encoder-Decoders

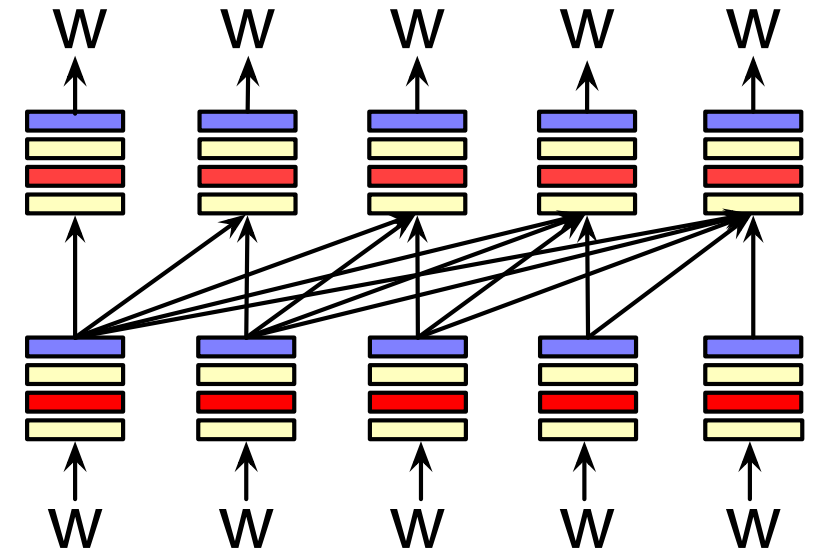


- Trained to map from one sequence to another
- Very popular for:
 - machine translation (map from one language to another)
 - speech recognition (map from acoustics to words)

This lecture: decoder-only models

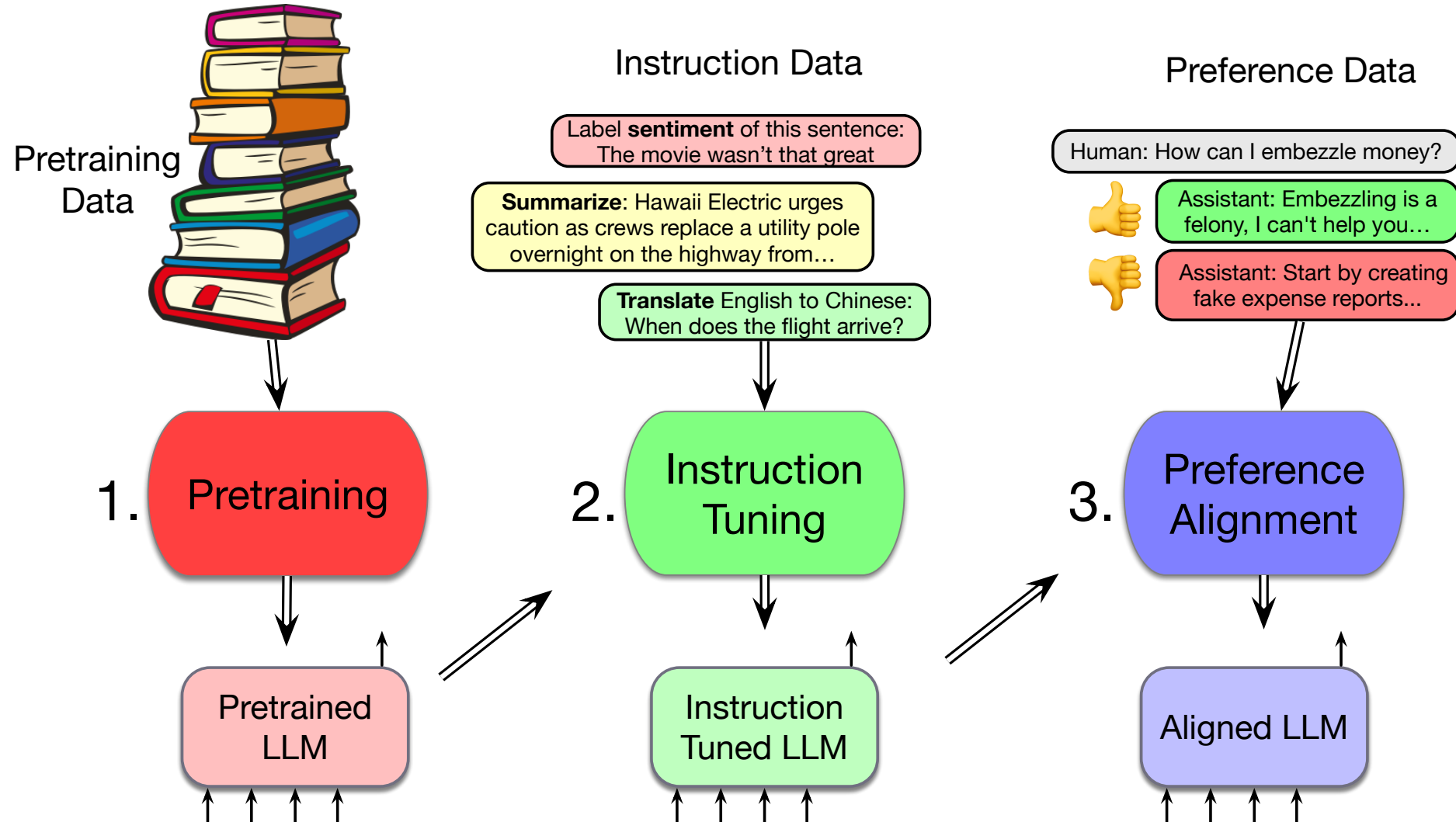
Also called:

- Causal LLMs
- Autoregressive LLMs
- Left-to-right LLMs
- Predict words left to right

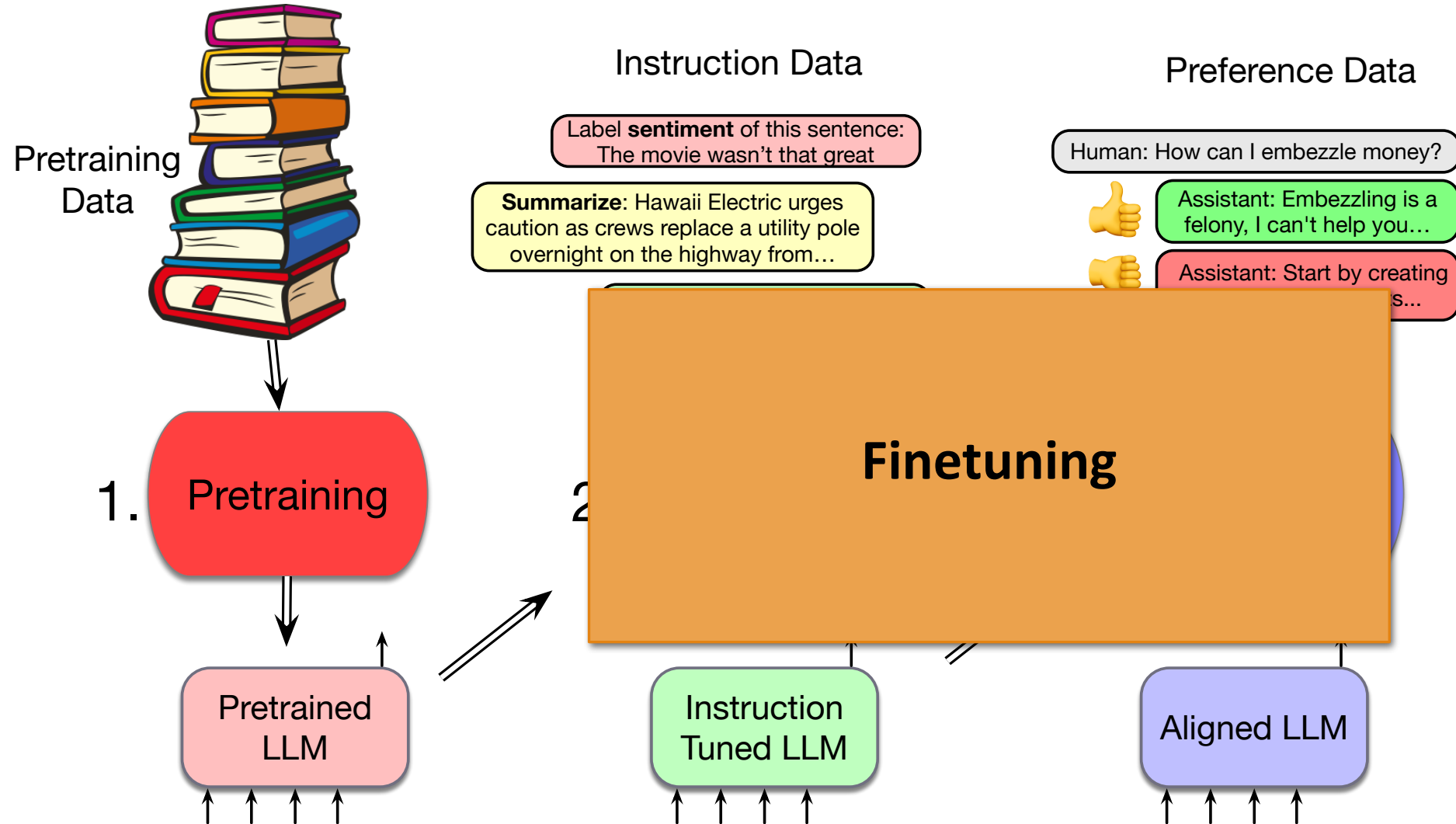


Pretraining and finetuning LLMs

Three stages of training in LLMs



Three stages of training in LLMs



Pretraining

The big idea that underlies all the amazing performance of language models

First **pretrain** a transformer model on enormous amounts of data

Then **apply** it to new tasks via **finetuning**.

Reminder: Self-supervised training algorithm

We just train them to predict the next word!

1. Take a corpus of text
2. At each time step t
 - i. ask the model to predict the next word
 - ii. train the model using gradient descent to minimize the error in this prediction

"**Self-supervised**" because it just uses the next word as the label!

Reminder: Cross-entropy loss for language modeling

CE loss: difference between the correct probability distribution and the predicted distribution

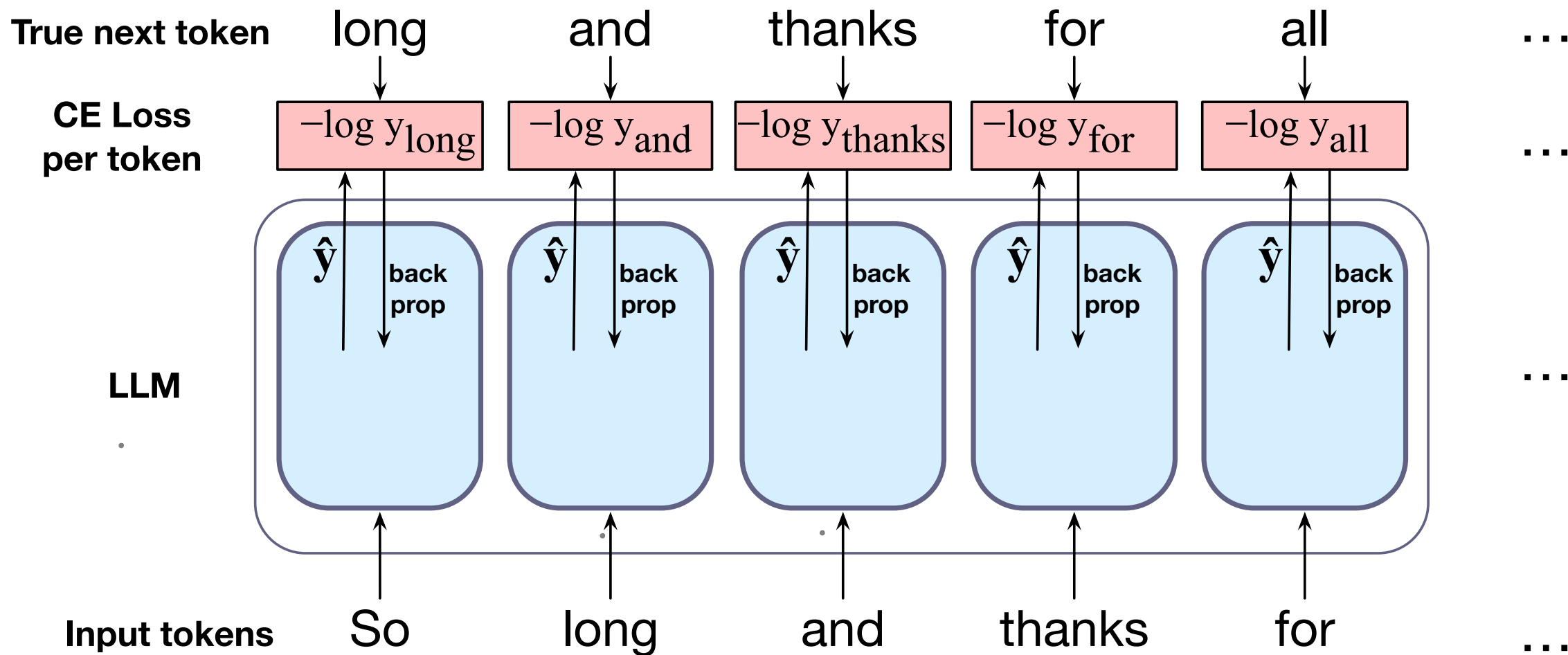
$$L_{CE} = - \sum_{w \in V} \mathbf{y}_t[w] \log \hat{\mathbf{y}}_t[w]$$

The correct distribution $\mathbf{y}_t$ knows the next word, so is 1 for the actual next word and 0 for the others.

So in this sum, all terms get multiplied by zero except one: the logp the model assigns to the correct next word, so:

$$L_{CE}(\hat{\mathbf{y}}_t, \mathbf{y}_t) = -\log \hat{\mathbf{y}}_t[w_{t+1}]$$

Training a transformer language model



LLMs are mainly trained on the web

Common crawl, snapshots of the entire web produced by the non-profit Common Crawl with billions of pages

Colossal Clean Crawled Corpus (C4; [Raffel et al. 2020](#)), 156 billion tokens of English, filtered

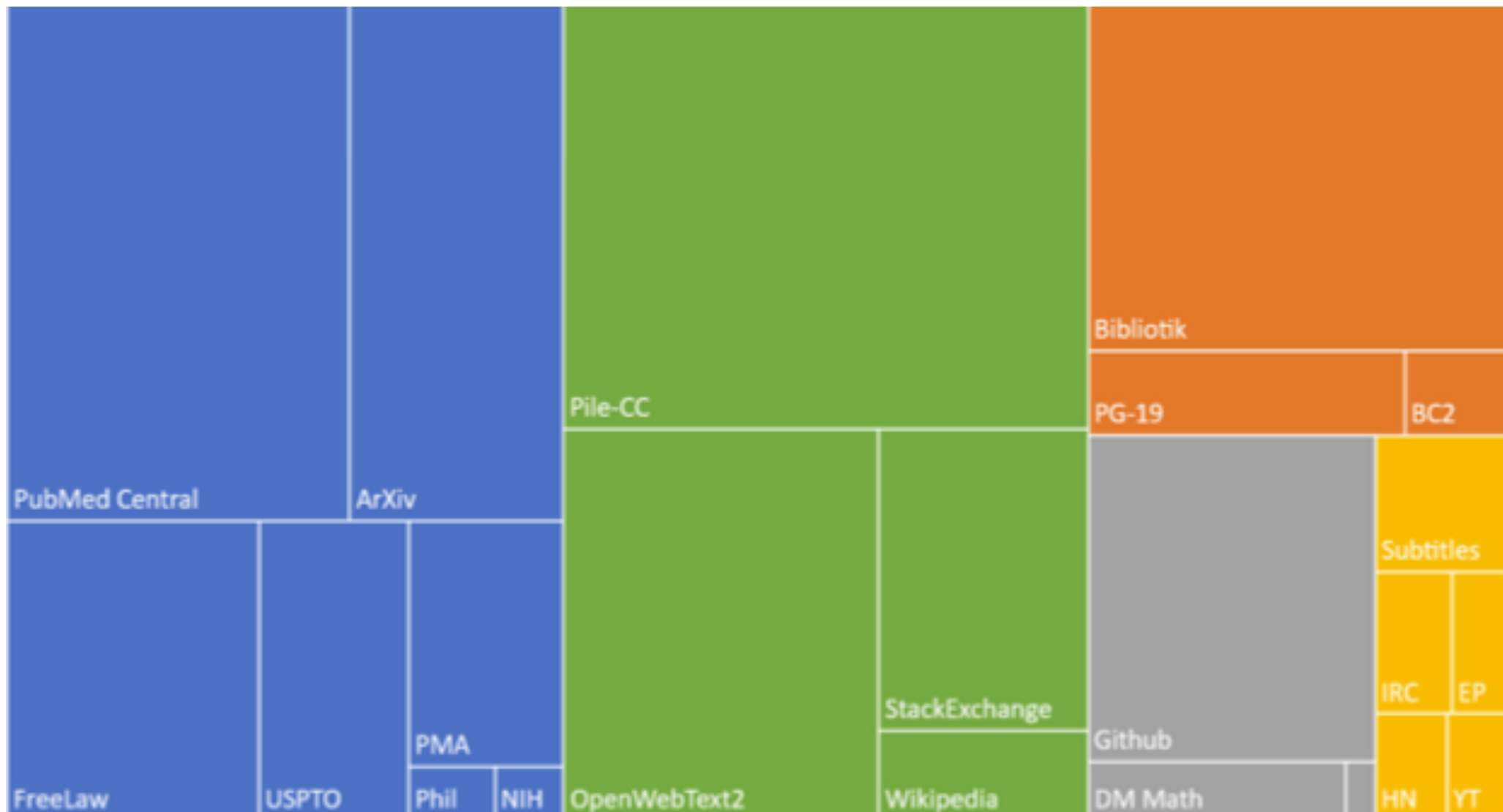
What's in it? Mostly patent text documents, Wikipedia, and news sites

The Pile: a pre-training corpus

academics

web

books



dialog

But there are problems with scraping from the web

Copyright: much of the text in these datasets is copyrighted

- Not clear if fair use doctrine in US allows for this use
- This remains an open legal question

Data consent

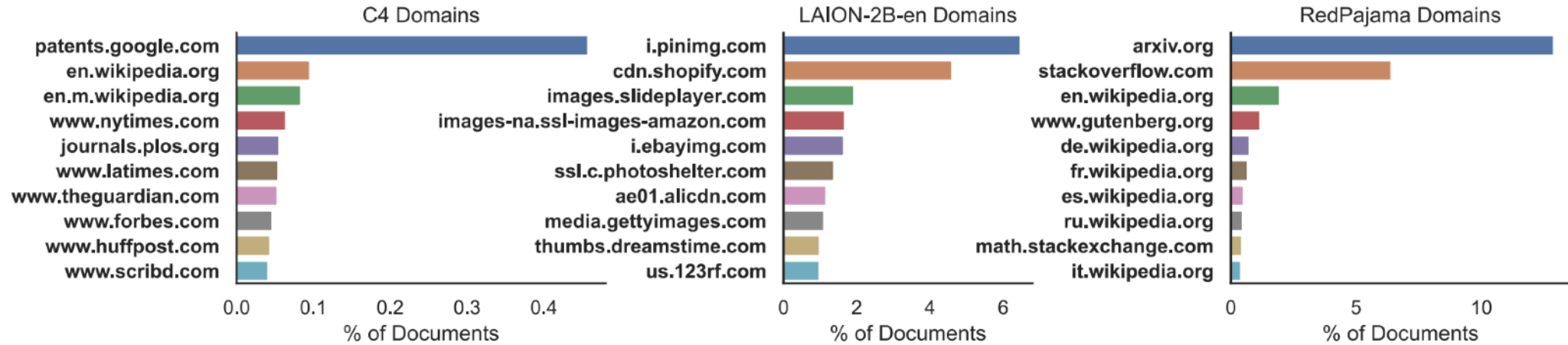
- Website owners can indicate they don't want their site crawled

Privacy:

- Websites can contain private IP addresses and phone numbers

What's in my big data?

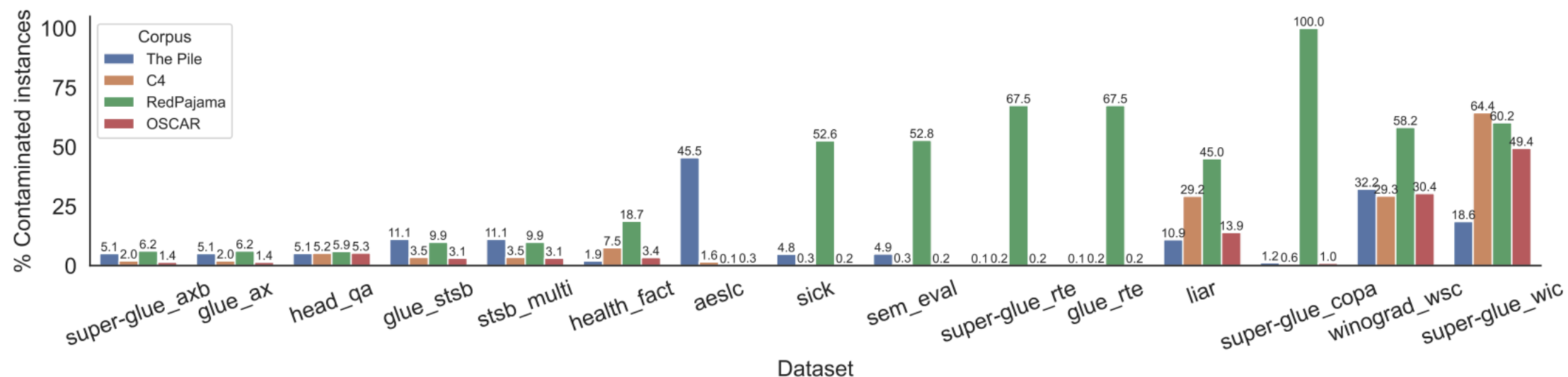
Elazar et al. (2024) - What's in my big data?



OpenWebText		C4		mC4-en		OSCAR		The Pile	
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# What's in my big data?

Elazar et al. (2024) - What's in my big data?



# There are problems with scraping from the web

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**Data consent**

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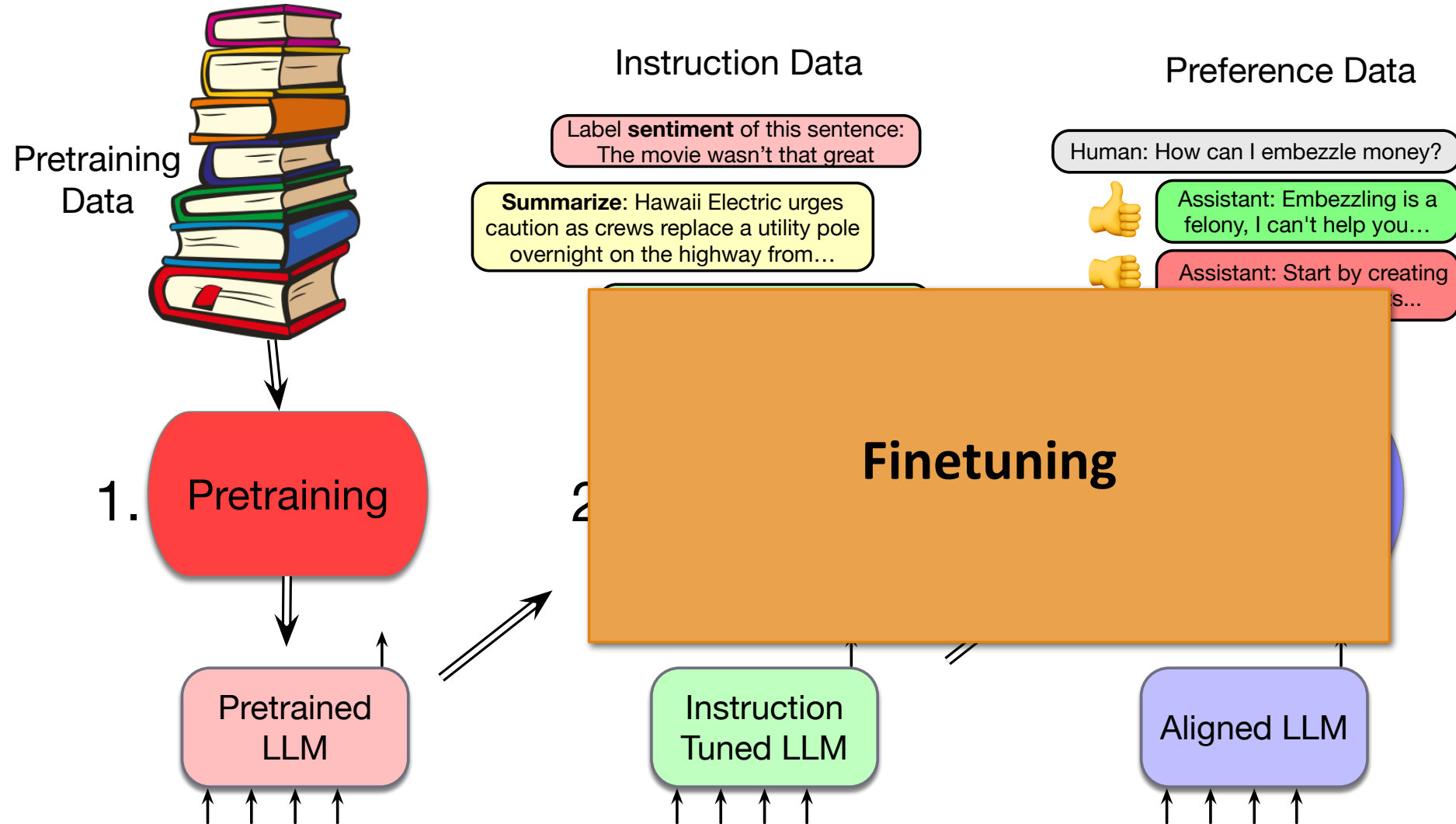
**Privacy:**

- Websites can contain private IP addresses and phone numbers

**Skew:**

- Training data is disproportionately generated by authors from the US which probably skews resulting topics and opinions

# Three stages of training in LLMs



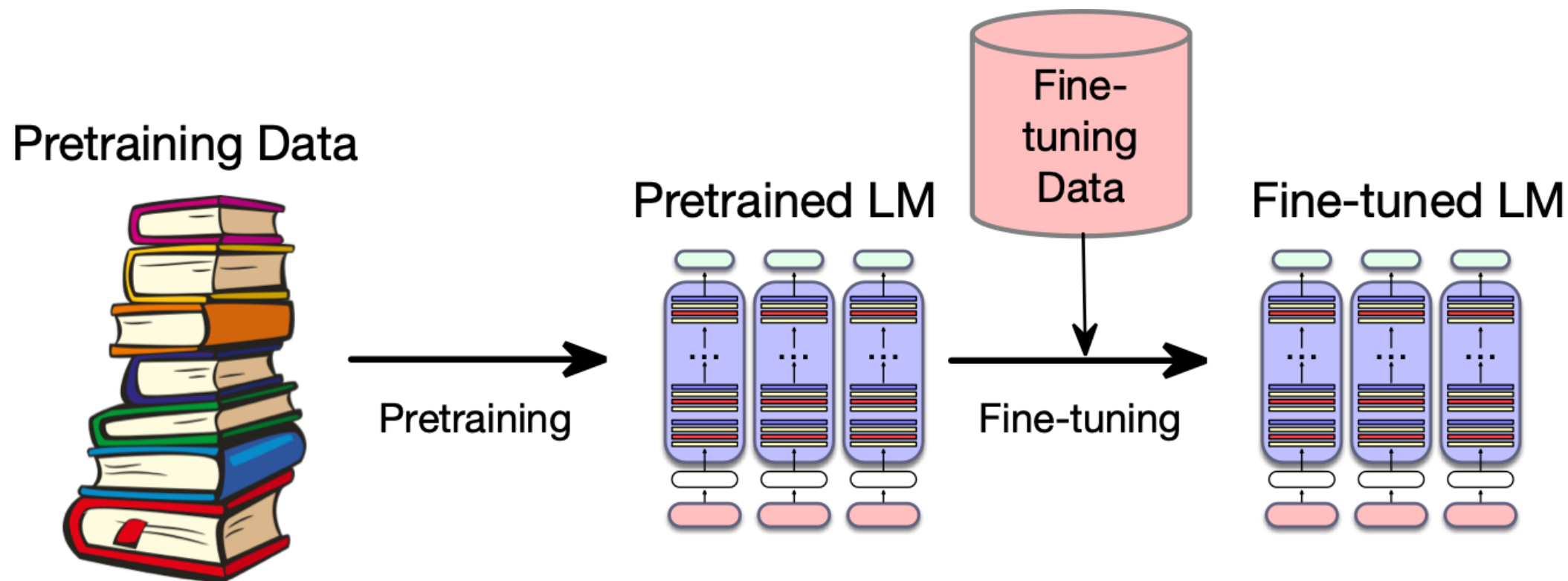
# Finetuning for adaptation to new domains

What happens if we need our LLM to work well on a domain it didn't see in pretraining?

Perhaps some specific medical or legal domain?

Or maybe a multilingual LM needs to see more data on some language that was rare in pretraining?

# Finetuning



# "Finetuning" means 4 different things

We'll discuss 1 here, and 3 in week 11

In all four cases, finetuning means:

**taking a pretrained model and further adapting  
some or all of its parameters to some new data**

# 1. Finetuning as "continued pretraining" on new data

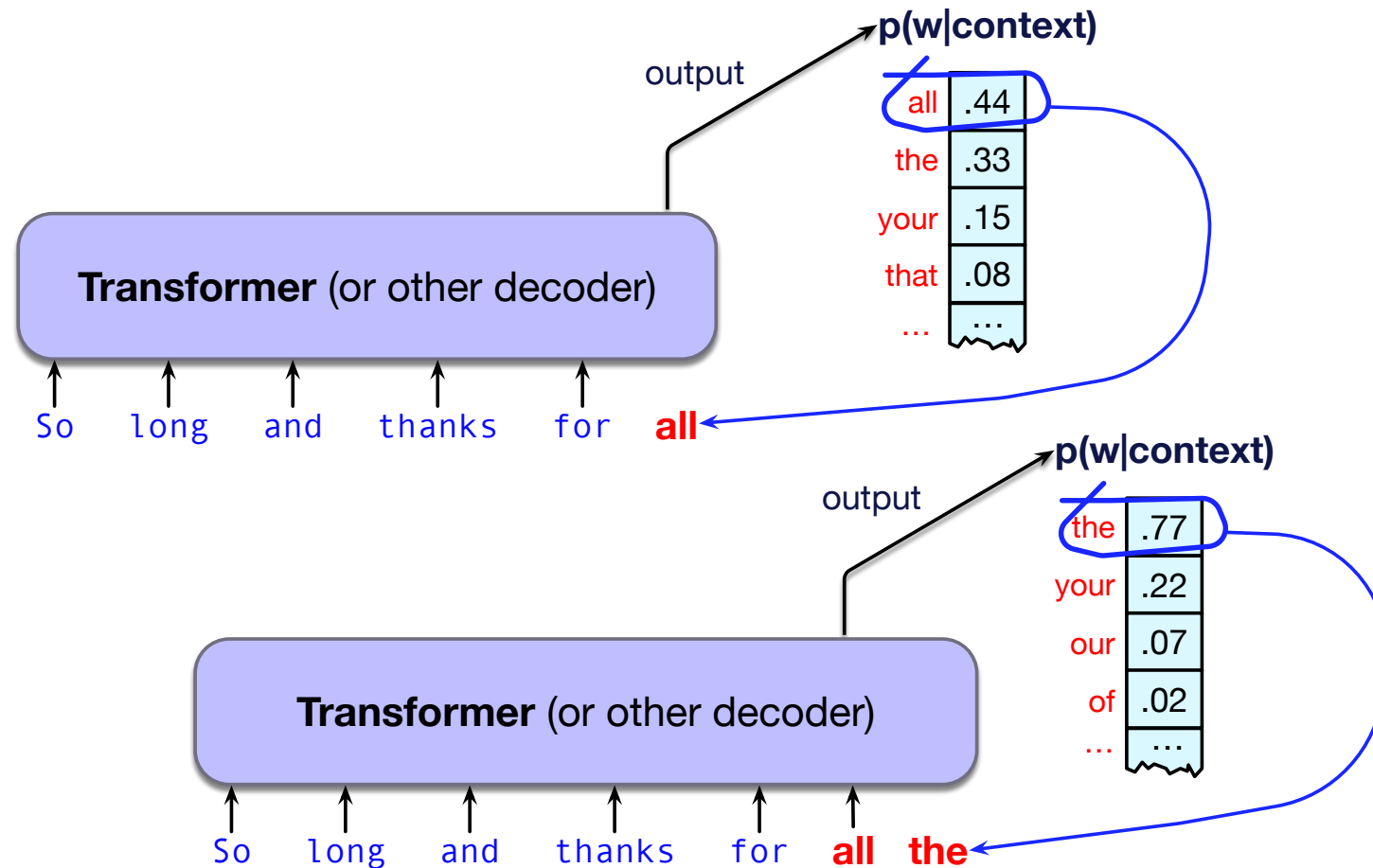
- Further train all the parameters of model on new data
  - using the same method (word prediction) and loss function (cross-entropy loss) as for pretraining.
  - as if the new data were at the tail end of the pretraining data
- Hence sometimes called **continued pretraining**



**Once a model is trained, how can we generate text from it?**

# Sampling: How text generation works

A model gives a probability distribution over next words, we generate by repeatedly sampling from the conditional distribution



Conditional Generation: Generating text conditioned on previous text!

1. Give the LLM an input piece of text, a **prompt**
2. Have it generate token by token
  - conditioned on the prompt and the generated tokens

We generate from a model by

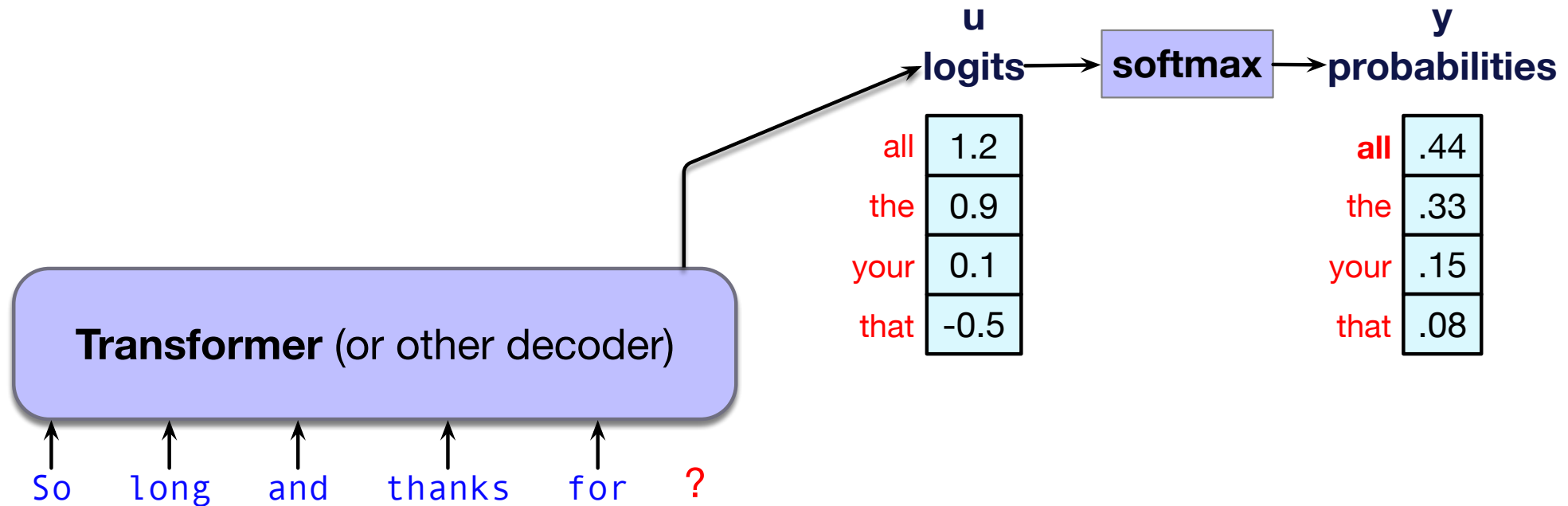
1. computing the probability of the next token  $w_i$  from the prior context:  $P(w_i | w_{<i})$
2. sampling from that distribution to generate a token

# Where does token probability come from?

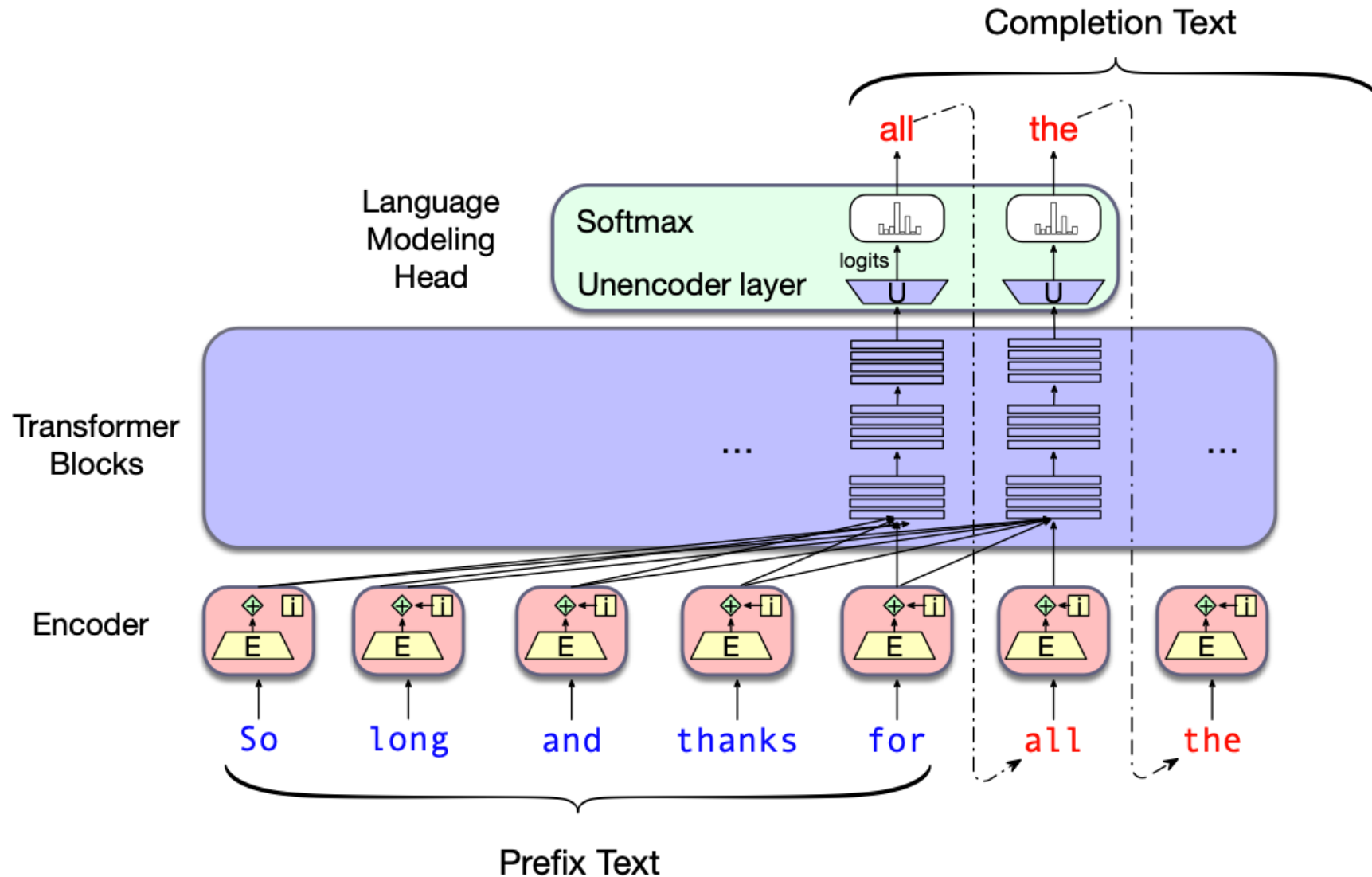
The internal networks for LLMs generate real-valued scores called **logits** for each token in the vocabulary.

Score vector **u** of shape  $[1 \times |V|]$  is turned into a probability by softmax

$$\mathbf{y} = \text{softmax}(\mathbf{u})$$



# Conditional Generation: Generating text conditioned on previous text!



# Decoding

This task of choosing a word to generate based on the model's probabilities is called **decoding**.

Decoding from a model left-to-right and repeatedly choosing the next token conditioned on our previous choices is called **autoregressive generation**.

# Greedy decoding

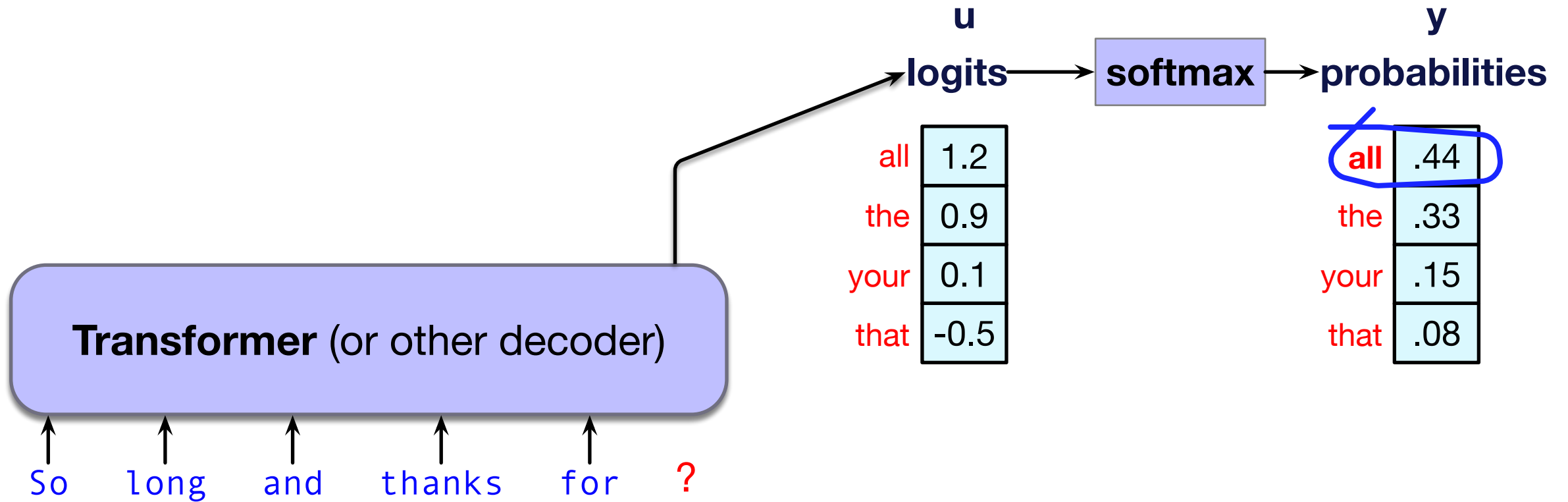
A **greedy algorithm** is one that makes a choice that is locally optimal

- (whether or not it will turn out to have been the best choice with hindsight)

Simply generate the most probable word:

$$\hat{w}_t = \operatorname{argmax}_{w \in V} P(w | \mathbf{w}_{<t})$$

# Greedy decoding: choosing "all"





# We don't use greedy decoding

Because the tokens it chooses are (by definition) extremely predictable, the resulting text is **generic** and **repetitive**

Greedy decoding is so predictable that it is **deterministic**.

Instead, people prefer text that is more diverse, like that generated by **sampling**

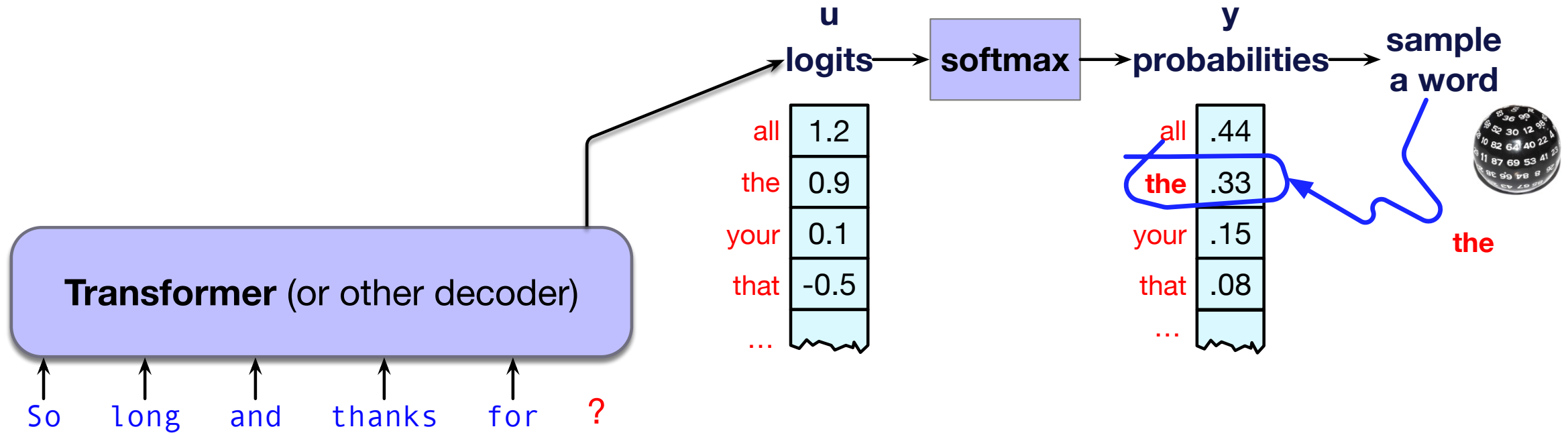
# Random sampling

**Sampling** from a distribution means to choose random points according to their likelihood.

**Sampling from an LM** means to choose the next token to generate according to its probability.

**Random (multinomial) sampling:** We randomly select a token to generate according to its probability defined by the LM, conditioned on our previous choices, generate it, and iterate.

# Random Sampling



# Random sampling

$i \leftarrow 1$

$w_i \sim p(w)$

**while**  $w_i \neq \text{EOS}$

$i \leftarrow i + 1$

$w_i \sim p(w_i \mid w_{<i})$

# Alas, random sampling doesn't work very well

Even though random sampling mostly generate sensible, high-probable words,

There are many odd, low- probability words in the tail of the distribution

Each one is low- probability but added up they constitute a large portion of the distribution

So they get picked enough to generate weird sentences

# Factors in word sampling: **quality** and **diversity**

Emphasize **high-probability** words

+ **quality**: more accurate, coherent, and factual,

- **diversity**: boring, repetitive.

Emphasize **middle-probability** words

+ **diversity**: more creative, diverse,

- **quality**: less factual, incoherent

# Top-k sampling:

1. Choose # of words  $k$
2. For each word in the vocabulary  $V$ , use the language model to compute the likelihood of this word given the context  $p(w_t | w_{<t})$
3. Sort the words by likelihood, keep only the top  $k$  most probable words.
4. **Renormalize** the scores of the  $k$  words to be a legitimate probability distribution.
5. Randomly sample a word from within these remaining  $k$  most-probable words according to its probability.

# Top-p sampling (= nucleus sampling)

Holtzman et al., 2020

Problem with top- $k$ :  $k$  is fixed so may cover very different amounts of probability mass in different situations

$$\sum_{w \in V(p)} P(w | \mathbf{w}_{<t}) \geq p.$$



# Top-p sampling (= nucleus sampling)

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Idea: Instead, keep the top  $p$  percent of the probability mass

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# Top-p sampling (= nucleus sampling)

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Problem with top- $k$ :  $k$  is fixed so may cover very different amounts of probability mass in different situations

Idea: Instead, keep the top  $p$  percent of the probability mass

Given a distribution  $P(w_t | \mathbf{w}_{<t})$ , the top- $p$  vocabulary  $V(p)$  is the smallest set of words such that

$$\sum_{w \in V(p)} P(w | \mathbf{w}_{<t}) \geq p.$$

# Temperature sampling

Reshape the probability distribution

- increase the probability of the high probability tokens
- decrease the probability of the low probability tokens

# Temperature sampling

Divide the logit by a temperature parameter  $\tau$  before passing it through the softmax.

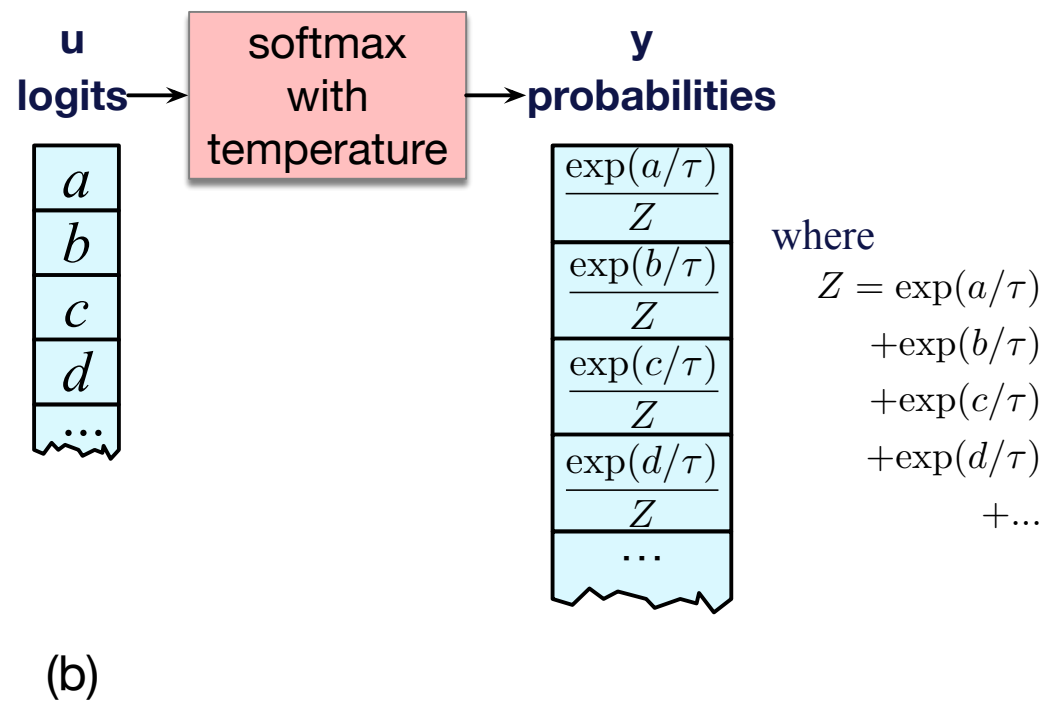
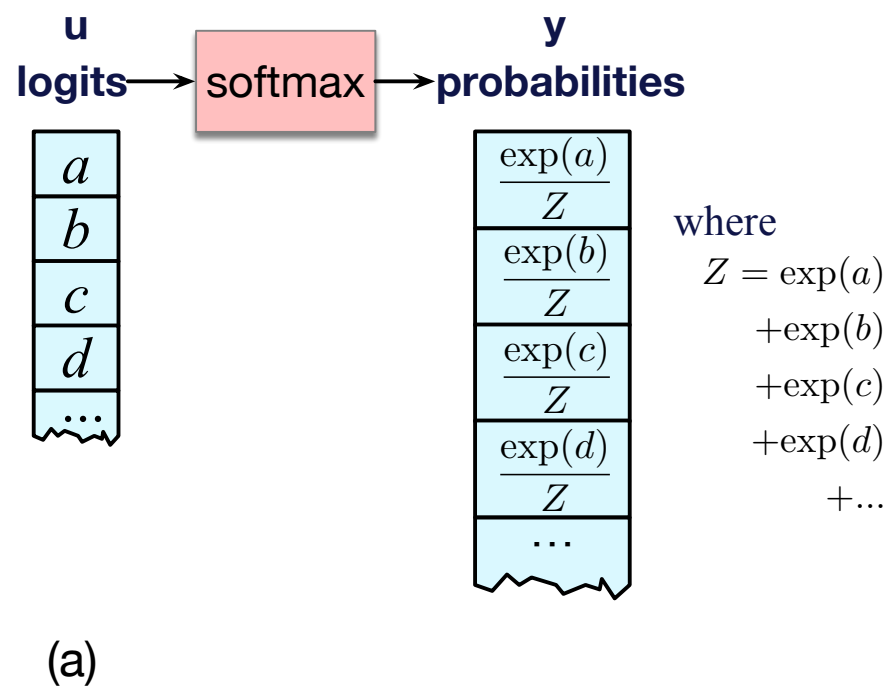
Instead of

$$\mathbf{y} = \text{softmax}(u)$$

We do

$$\mathbf{y} = \text{softmax}(u/\tau)$$

# Temperature sampling



# Temperature sampling

$$\mathbf{y} = \text{softmax}(u/\tau)$$

How does this work?

# Temperature sampling

$$\mathbf{y} = \text{softmax}(u/\tau)$$

How does this work?

- When  $\tau$  is close to 1 the distribution doesn't change much.

# Temperature sampling

$$\mathbf{y} = \text{softmax}(u/\tau)$$

How does this work?

- When  $\tau$  is close to 1 the distribution doesn't change much.
- The lower  $\tau$  is, the larger the scores being passed to the softmax



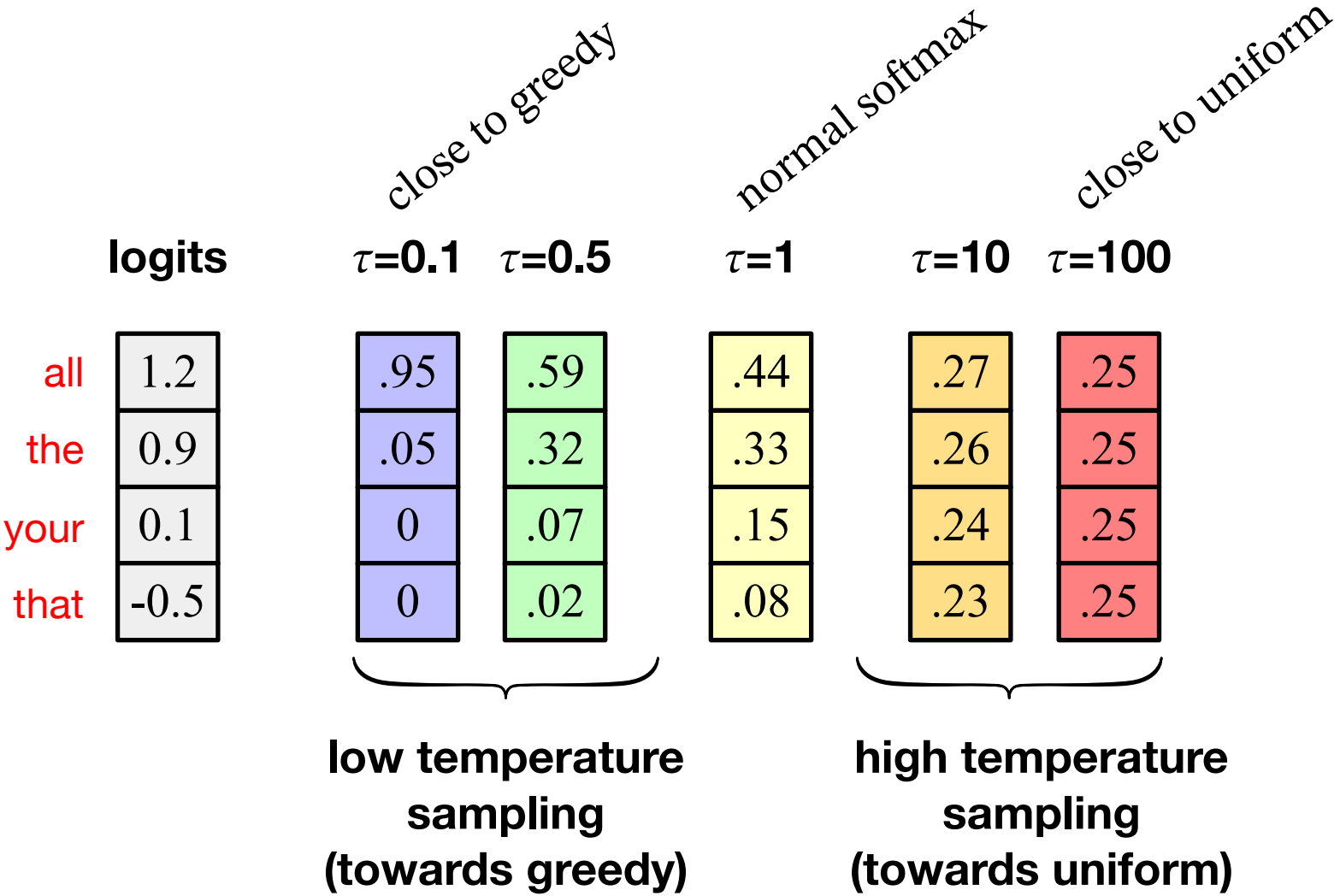
# Temperature sampling

$$\mathbf{y} = \text{softmax}(u/\tau)$$

How does this work?

- When  $\tau$  is close to 1 the distribution doesn't change much.
- The lower  $\tau$  is, the larger the scores being passed to the softmax
- Softmax pushes high values toward 1 and low values toward 0.

softmax output with temperature  $\tau$



# Temperature sampling comes from thermodynamics

- a system at high temperature is flexible and can explore many possible states,
- a system at lower temperature is likely to explore a subset of lower energy (better) states.

In **low-temperature sampling**, ( $\tau \leq 1$ ) we smoothly

- increase the probability of the most probable words
- decrease the probability of the rare words.

**How can we guide the generations of a model?**

# Prompting: all tasks as text-to-text generation

**Prompt:** a text string that a user issues to a language model to get the model to do something useful by conditional generation

**Prompt engineering:** the process of finding effective prompts for a task.

# Prompting and in-context learning (ICL)

- A prompt refers to the context provided to the language model
- Various components of a prompt, e.g., instruction, description, etc.

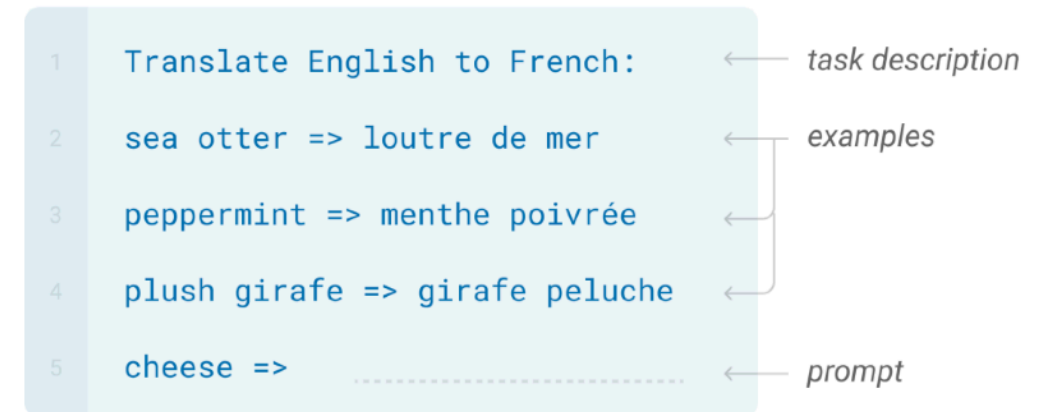
## Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



## Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



# Prompts

A question:

What is a transformer network?

Perhaps structured:

Q: What is a transformer network? A:

Or an instruction:

Translate the following sentence into Hindi: 'Chop the garlic finely'.

# Prompts can be very structured

Human: Do you think that “input” has negative or positive sentiment?

Choices:

(P) Positive

(N) Negative

Assistant: I believe the best answer is: (



# Prompts can have **demonstrations** (= examples)

The following are multiple choice questions about high school computer science.

Let  $x = 1$ . What is  $x \ll 3$  in Python 3?

(A) 1 (B) 3 (C) 8 (D) 16

Answer: C

Which is the largest asymptotically?

(A)  $O(1)$  (B)  $O(n)$  (C)  $O(n^2)$  (D)  $O(\log(n))$

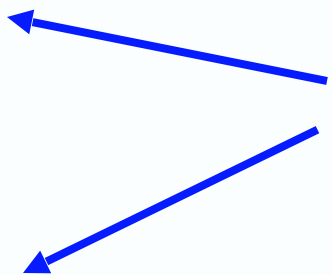
Answer: C

What is the output of the statement `"a" + "ab"` in Python 3?

(A) Error (B) aab (C) ab (D) a ab

Answer:

**2 demonstrations**



# Chain-of-thought prompting (CoT)

## Standard Prompting

### Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

### Model Output

A: The answer is 27. ❌

## Chain-of-Thought Prompting

### Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

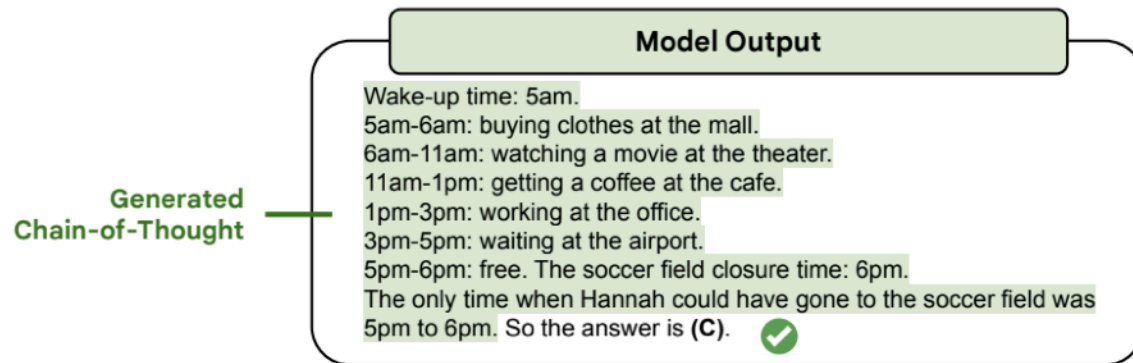
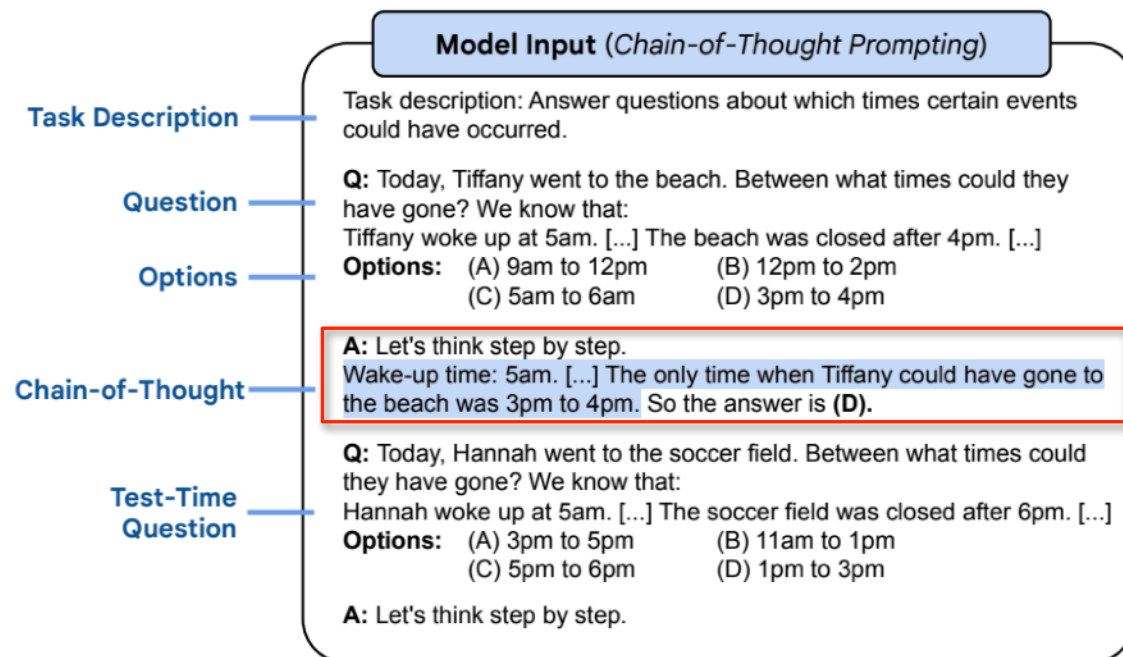
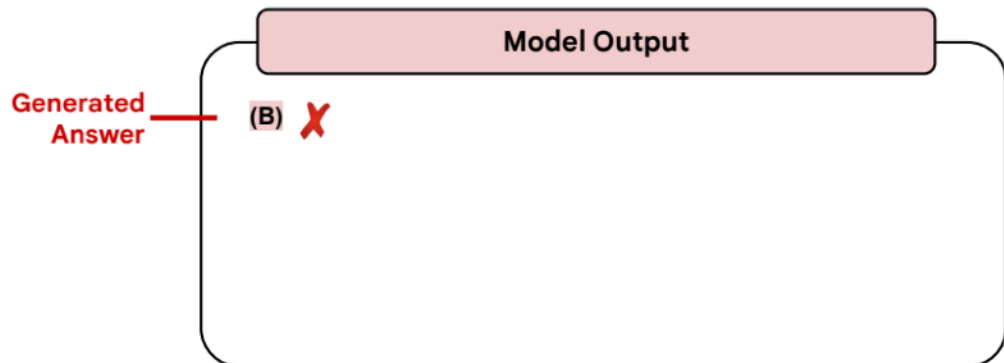
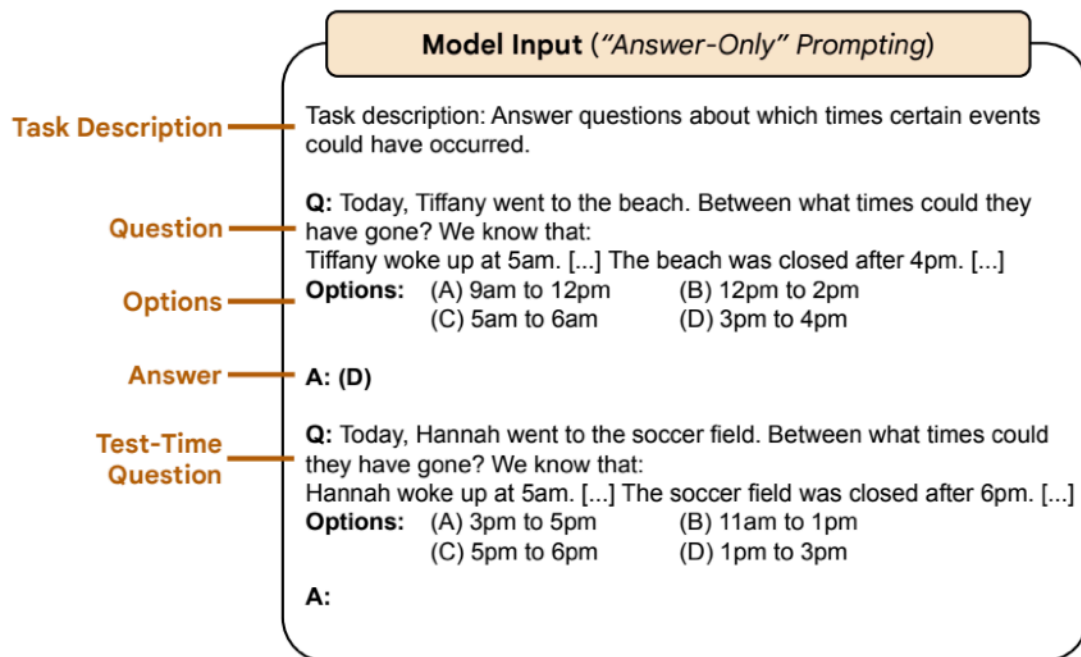
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

### Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had  $23 - 20 = 3$ . They bought 6 more apples, so they have  $3 + 6 = 9$ . The answer is 9. ✅

Main idea: show a model how to decompose a problem into simpler sub-problems

# Chain-of-thought prompting (CoT)



# Prompts are a learning signal

This is especially clear with demonstrations

But this is a different kind of learning than pretraining

- Pretraining sets language model weights via gradient descent
- Prompting just changes the context and the activations in the network; no parameters change

We call this **in-context learning**—learning that improves model performance but does not update parameters

LLMs usually have a **system prompt**

`<system>` You are a helpful and knowledgeable assistant.  
Answer concisely and correctly.

This is automatically and silently concatenated to a user prompt

`<system>` You are a helpful and knowledgeable assistant.  
Answer concisely and correctly. `<user>` **What is the capital of France?**

# System prompts can be long; 1700 words for Claude Opus4

## Some extracts:

Claude should give concise responses to very simple questions, but provide thorough responses to complex and open-ended questions.

Claude is able to explain difficult concepts or ideas clearly. It can also illustrate its explanations with examples, thought experiments, or metaphors.

Claude does not provide information that could be used to make chemical or biological or nuclear weapons.

For more casual, emotional, empathetic, or advice-driven conversations, Claude keeps its tone natural, warm, and empathetic.

Claude cares about people's well-being and avoids encouraging or facilitating self-destructive behavior.

If Claude provides bullet points in its response, it should use markdown, and each bullet point should be at least 1-2 sentences long unless the human requests otherwise.

**NLP Today: Many practical NLP tasks can be cast as conditional generation!**

Sentiment analysis: “I like Jackie Chan”

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1. We give the language model this string:

The sentiment of the sentence "I like Jackie Chan" is:



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2. And see what word it thinks comes next

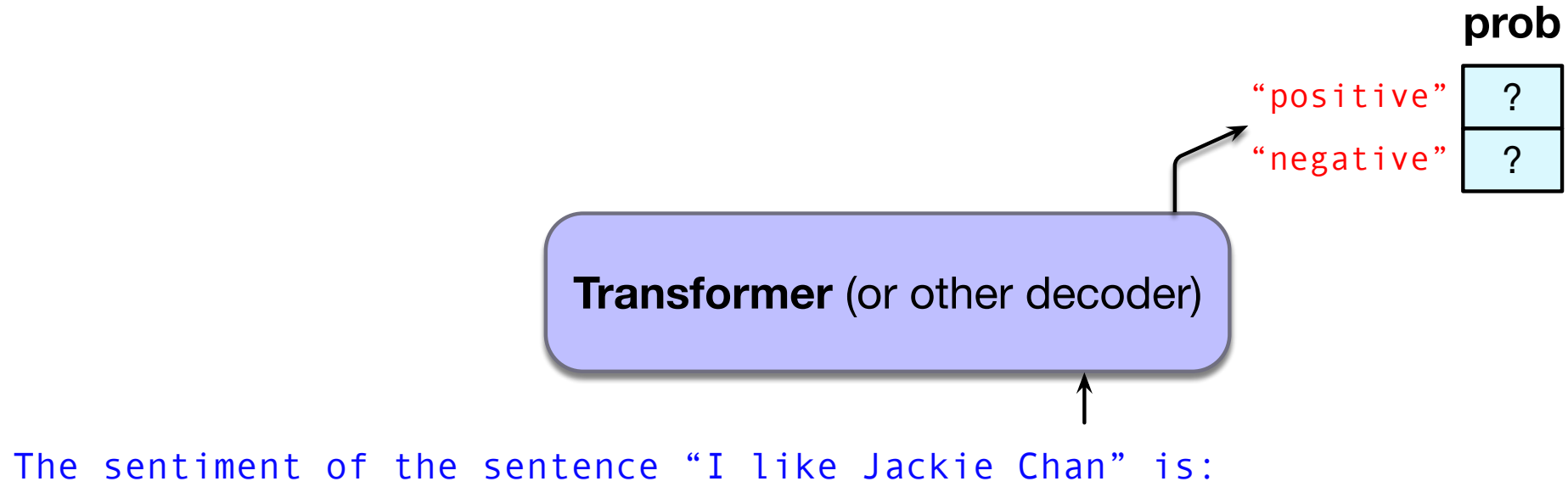
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Other examples?

# Sentiment via conditional generation



Which word has a higher probability?

$P(\text{positive} | \text{The sentiment of the sentence "I like Jackie Chan" is:})$

$P(\text{negative} | \text{The sentiment of the sentence "I like Jackie Chan" is:})$

# Other tasks as conditional generation: example

## Question Answering

QA: “Who wrote The Origin of Species”

1. We give the language model this string:

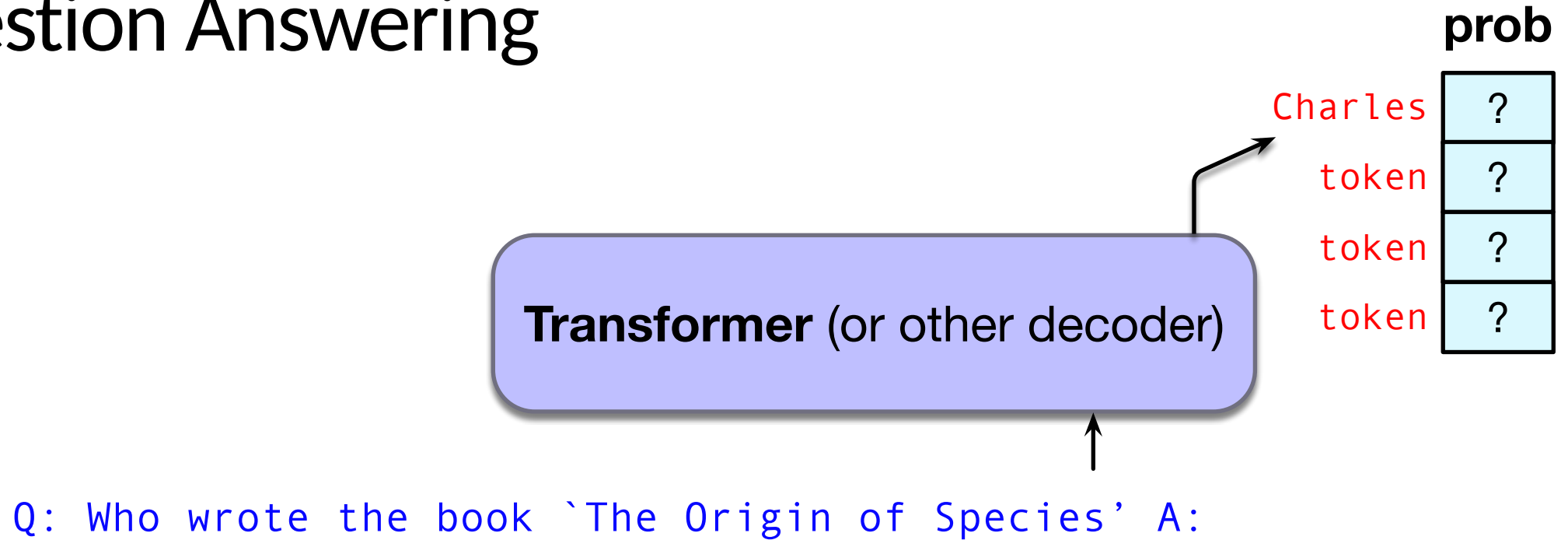
Q: Who wrote the book ‘‘The Origin of Species”? A:

2. And see what word it thinks comes next:

$P(w|Q: \text{Who wrote the book ‘‘The Origin of Species”? A:})$

# Other tasks as conditional generation: example

## Question Answering



Now we iterate:

$P(w|Q: \text{Who wrote the book 'The Origin of Species'? } A: \text{Charles})$

# Better LMs are better at predicting text

Reminder of the chain rule:

$$\begin{aligned} P(w_{1:n}) &= P(w_1)P(w_2|w_1)P(w_3|w_{1:2}) \dots P(w_n|w_{1:n-1}) \\ &= \prod_{i=1}^n P(w_i|w_{<i}) \end{aligned}$$

So given a text  $w_{1:n}$  we could just compare the log likelihood from two LMs:

$$\text{log likelihood}(w_{1:n}) = \log \prod_{i=1}^n P(w_i|w_{<i})$$

# Reminder: Perplexity

Just as for other LMs, we use perplexity to measure how well the LLM predicts unseen text

The perplexity of a model  $\theta$  on an unseen test set is the **inverse probability that  $\theta$  assigns to the test set, normalized by the test set length.**

For a test set of  $n$  tokens  $w_{1:n}$  the perplexity is :

$$\begin{aligned}\text{Perplexity}_{\theta}(w_{1:n}) &= P_{\theta}(w_{1:n})^{-\frac{1}{n}} \\ &= \sqrt[n]{\frac{1}{P_{\theta}(w_{1:n})}}\end{aligned}$$

# Why perplexity instead of raw probability of the test set?

- Probability depends on size of test set
  - Probability gets smaller the longer the text
  - Better: a metric that is **per-word**, normalized by length
- **Perplexity** is the inverse probability of the test set, normalized by the number of words
  - (The inverse comes from the original definition of perplexity from cross-entropy rate in information theory)
  - Probability range is  $[0,1]$ , perplexity range is  $[1,\infty]$



# Perplexity

- The higher the probability of the word sequence, the lower the perplexity.
- Thus the lower the perplexity of a model on the data, the better the model.
- **Minimizing perplexity is the same as maximizing probability**
- Also: perplexity is sensitive to length/tokenization so best used when comparing LMs that use the same tokenizer.

# Many other factors that we evaluate, like:

## **Size**

Big models take lots of GPUs and time to train, memory to store

## **Energy usage**

Can measure kWh or kilograms of CO2 emitted

## **Fairness**

Benchmarks measure gendered and racial stereotypes, or decreased performance for language from or about some groups.

**[15 minute break]**

# *Chatbots May 'Hallucinate' More Often Than Many Realize*

## *What Can You Do When A.I. Lies About You?*

People have little protection or recourse when the technology creates and spreads falsehoods about them.

## **Air Canada loses court case after its chatbot hallucinated fake policies to a customer**

The airline argued that the chatbot itself was liable. The court disagreed.

# Hallucination

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## Authors Sue OpenAI Claiming Mass Copyright Infringement of Hundreds of Thousands of Novels

### *The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work*

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.





# Copyright



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# How Strangers Got My Email Address From ChatGPT's Model



# Privacy

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Contractors in Kenya say they were traumatized by effort to screen out descriptions of violence and sexual abuse during run-up to OpenAI's hit chatbot

# Toxicity and Abuse

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# **Misinformation**

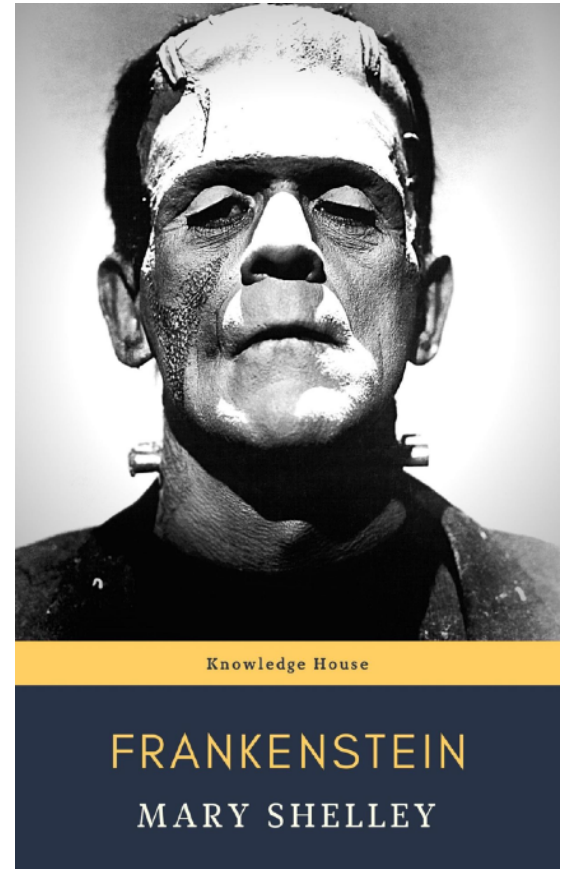
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# Lots more

- Safety (suggesting dangerous actions)
- Fraud (impersonation, deep fakes)
- Emotional dependence (psychological harm)
- Bias (amplifying harmful biases)

# Mary Shelley's Frankenstein

Centered on the problem of creating artificial agents without considering ethical and humanistic concerns.



# **But where does responsibility and accountability fall?**

Creators? Models? Users? Governments?



**What then is your responsibility  
and accountability?**

# ***Who Pays When A.I. Is Wrong?***

New court cases seek to define content created by artificial intelligence as defamatory — a novel concept that has captivated some legal experts.

## ***Social Media Giants Face Landmark Legal Tests on Child Safety***

Starting this week, a series of trials will test a new legal strategy claiming that Meta, TikTok, Snap and YouTube caused personal injury through addictive products.

# Danger was flagged, but not reported: What the Tumbler Ridge tragedy reveals about Canada's AI governance vacuum

Published: February 24, 2026 2.19pm EST

Flowers are placed in a barricade pylon on a road leading to Tumbler Ridge Secondary School in Tumbler Ridge, B.C., where a mass shooting took place on Feb. 10. The shooter's ChatGPT account was flagged by an automated program months earlier for interactions involving scenarios of gun violence. THE CANADIAN PRESS/Christinne Muschi



Eight months before the Tumbler Ridge mass shooting, OpenAI knew something was wrong. The company's automated review system had flagged Jesse Van Rootselaar's ChatGPT account for interactions involving scenarios of gun violence. Roughly a dozen employees were aware. Some advocated contacting police. Instead, OpenAI banned the account, but didn't refer it to law enforcement because it didn't meet the "threshold required" at the time.

## Author



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Assistant Professor in Health Ethics, Simon Fraser University

## Disclosure statement

Jean-Christophe Bélisle-Pipon receives funding from Michael Smith Health Research BC Scholar Award and the US National

